

Here, the return on investment is:  $\$5612 \div \$20,000 = 28.06\%$  which, when compared to the alternate bet, patently reflects an incorrect call. Again, making this call makes or breaks the arbitrage opportunity.

A simple model of evaluating completion provides the market's assessment of the likelihood (not to be confused with desirability) of completion. Brown and Raymond (1986) provide a formal model for the prediction of successful mergers and acquisitions. Given an announcement of consideration, if the merger is successful a long-short position would return a yield:

$$(P_{Tt} - P_{mt})/P_{mt} \quad (13.28)$$

However, as the tender is not given to be accepted, the risk of falling through must be considered through the possibility of a negative return:

$$(P_F - P_{mt})/P_{mt} \quad (13.29)$$

where  $P_F$  is the price the target stock falls to if the merger is unsuccessful. Indeed it cannot be said with certainty what value  $P_F$  will take. However, pre-announcement prices can be used as proxies to assess the overall risk-reward profile.

Given this framework, we can assess the market's prediction of the likelihood of the merger. Computing the period  $t$  merger probability  $x_t$  given the merger arbitrage payoff is zero:

$$\begin{aligned} E(\Pi_t) &= x_t[(P_{Tt} - P_{mt})/P_{mt}] + (1 - x_t)[(P_F - P_{mt})/P_{mt}] \\ &= x_t[(P_{Tt}/P_{mt}) - 1] + (1 - x_t)[(P_F/P_{mt}) - 1] \\ &= 0 \end{aligned} \quad (13.30)$$

Thus:

$$x_t = \frac{1 - (P_F/P_{mt})}{(P_{Tt}/P_{mt}) - (P_F/P_{mt})} \quad (13.31)$$

where  $(P_{Tt}/P_{mt})$  and  $(P_F/P_{mt})$  are defined as tender and failure premiums respectively. Thus likelihood  $x_t$  describes the price assimilation of the tender into the market price as of period  $t$ .

Merger arbitrage is deal driven and not market driven. Consequently returns are correlated to the relative value between the two stocks when the arbitrage is executed and as such they are not, in general, correlated to returns of the general stock market (Anson, 2008). Exceptions to this rule are statistical speculators who will actively seek risk arbitrage opportunities.